Simulation and Modeling

Lecture 03

Simulation Development Example

Single Server Queueing System (SSQS)

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SSQS: Problem Statement

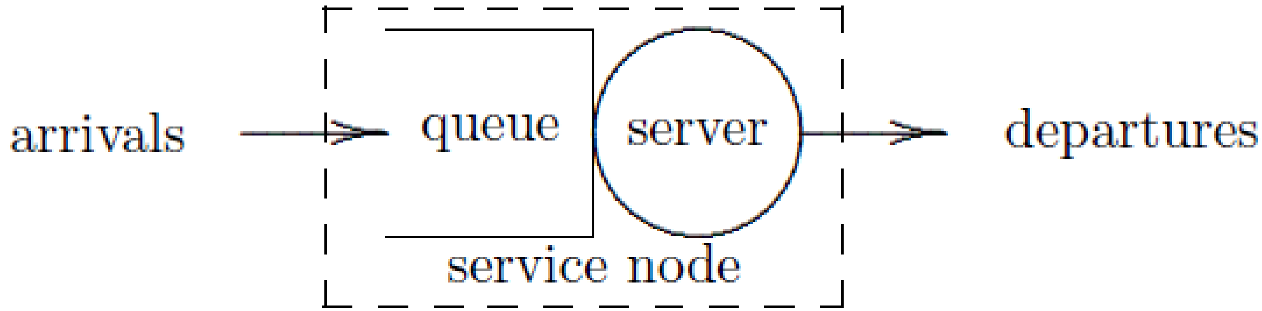
- Customers arrive at random time-points seeking service which requires a random time
- If arriving customer finds the server idle
  - Immediately enters service
  - Otherwise, enters the queue
- Quantity of interest for evaluation
  - Service quality improvement
  - Customer satisfaction
  - Effective use of system resources

SSQS – Step 1: Goals and Objectives

- Simple Boolean decision
  - Do we need an additional server
- Numeric Decision
  - How many parallel server do we need?
- Qualitative Decision
  - Do we need to decrease the customer waiting time in the queue

SSQS – Step 2: Conceptual Model (1/10)

1. System Diagram



Single Server Queueing System

SSQS – Step 2: Conceptual Model (3/10)

2. System State and State Variable (1/3)

- What an arriving customer will find?
  - Server is idle or busy
    - Server status
  - Long or small queue
    - Queue length
- System state may be represented by two variables – server status, queue length
- <u>Relationship of state variables</u>
  - If the queue length is non-zero, server must be busy
  - If the server is idle, queue length must be zero

SSQS – Step 2: Conceptual Model (6/10)

3. Events (2/2)

- System state changes only when customer arrives or leaves the system
  - Customer arrives (arrival event)
  - Service completed and customer leaves (departure event)
- System state (server status, queue length)
  - Arrival either changes the server status from idle to busy, or increases the queue length
  - Departure either changes the server status from busy to idle, or decreases the queue length

SSQS – Step 2: Conceptual Model (7/10)

4. State Space

- State variable (server status, queue length)
  - System state is a set of two element
  - X is (0,0), (1, 0), (1, 1), (1, 2), (1, 3), …

SSQS – Step 2: Conceptual Model (8/10)

5. Feasible event set at state x

- State variable (server status, queue length)
  - X is (0,0), (1, 0), (1, 1), (1, 2), (1, 3), …
  - At state (0,0): only arrival
  - Other states: both arrival and departure

SSQS – Step 3: Specification Model (2/14)

1. State Equations (1/2)

- Svariables are
  - $q(t)$– queue length at time t
  - $x(t)$ – status of the server, 0 – idle, 1 – busy
- Accordingly, the state equations are

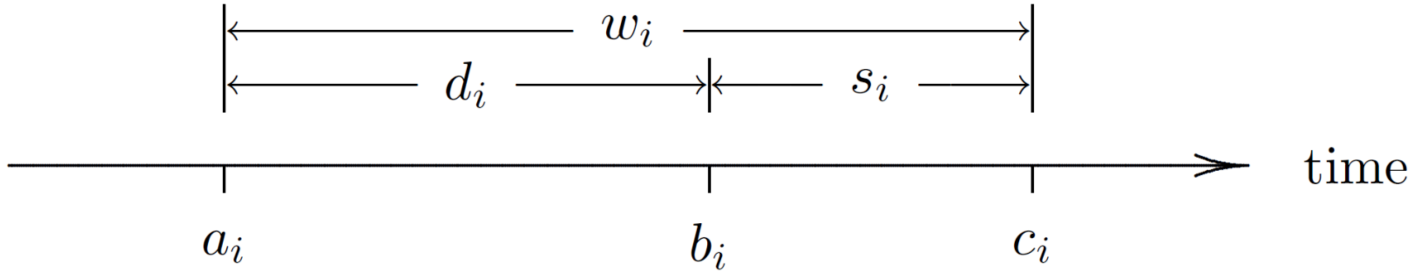
$$x(t^+) = \begin{cases} x(\mathrm{t}) == 0\ ?\ x(t)+1 : x(t) & \text{if arrival occurs at time } t \\ q(\mathrm{t}) == 0\ ?\ x(t)\text{-}1 : x(t) & \text{if departure occurs at time } t \\ x(t) & \text{otherwise} \end{cases}$$

$$q(t^+) = \begin{cases} x(t) > 0\ ?\ q(t)+1 : q(t) & \text{if arrival occurs at time } t \\ q(t) > 0\ ?\ q(t)-1) : q(t) & \text{if departure occurs at time } t \\ q(t) & \text{otherwise} \end{cases}$$

SSQS – Step 3: Specification Model (4/14)

4. Output Equations (1/5)

- A number of variables is associated with job i
  - The arrival time is a_i
  - The delay in the queue is $d_i \geq 0$
  - The service start time is $b_i = a_i + d_i$
  - The service time is $s_i > 0$
  - The system time is $w_i = d_i + s_i$
  - The departure (completion) time is $c_i = w_i + s_i$



SSQS – Step 3: Specification Model (9/14)

Output statistics – Job-averaged (2/6)

- Average interarrival time

$$\bar{r} = \frac{1}{n}\sum_{i=1}^{n} r_i = \frac{a_n}{n}$$

- Average service time

$$\bar{s} = \frac{1}{n}\sum_{i=1}^{n} s_i$$

- Average delay

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i$$

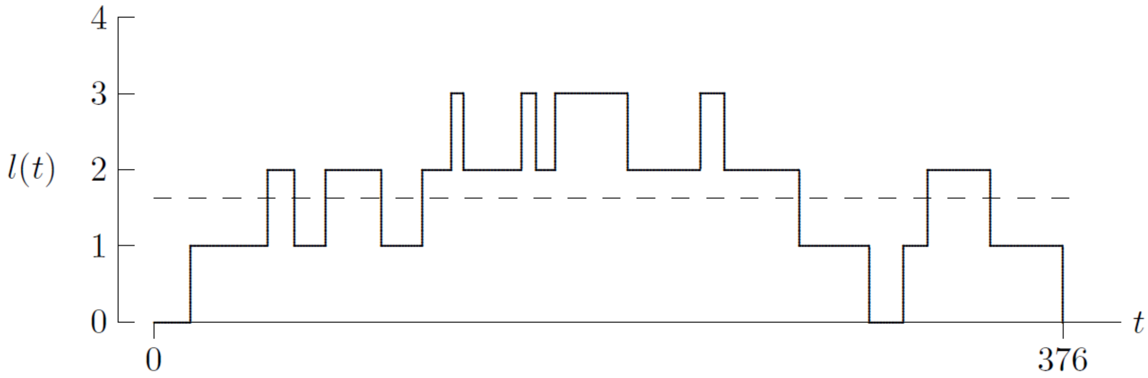
- Average system time

$$\bar{w} = \frac{1}{n}\sum_{i}^{n} w_i = \frac{1}{n}\sum_{i}^{n}(d_i + s_i)$$
$$= \frac{1}{n}\sum_{i=1}^{n} d_i + \frac{1}{n}\sum_{i=1}^{n} s_i = \bar{d} + \bar{s}$$

SSQS – Step 3: Specification Model (10/14)

Output statistics – Time-averaged (3/6)

- Three additional variables are
  - Number of jobs in the system at time t
    - $l(t) = 0,1,2,\ldots$
  - Number of jobs in the queue at time t
    - $q(t) = 0,1,2,\ldots$
  - Number of jobs in service at time t
    - $u(t) = 0,1$
  - By definition, $l(t) = q(t) + u(t)$



Time-averaged number in the system

SSQS – Step 3: Specification Model (11/14)

Output statistics – Time-averaged (4/6)

- The time-averaged number of job in the system for period $(0, \tau)$
$$\bar{l} = \frac{1}{\tau}\int_0^{\tau} l(t)dt$$
- The time-averaged number of job in the queue for period $(0, \tau)$
$$\bar{q} = \frac{1}{\tau}\int_0^{\tau} q(t)dt$$
- The Time-averaged number of job in service for period $(0, \tau)$
$$\bar{u} = \frac{1}{\tau}\int_0^{\tau} u(t)dt$$
  - This is also known as server utilization
    - Proportion of time the server is busy
- Since $l(t) = q(t) + u(t)$, for all $t > 0$, $\bar{l} = \bar{q} + \bar{u}$

SSQS – Step 3: Specification Model (14/14)

Input Data Modeling

- Job interarrival times
- Assumed to be exponential with parameter $\lambda = 1$ minute
  - PDF is $f_R(r) = \frac{1}{\lambda} e^{-\frac{r}{\lambda}}, \quad r \geq 0$
    - First a random number $\sim U(0,1)$ is generated
    - Then, a random variate $\sim Exp(\lambda)$ is generated
      - $R = -\lambda \ \ln U$
- Job service times
  - Also assumed as exponential with parameter $\beta = 0.5$ minute
    - $R = -\beta \ \ln U$
- Simulation time
  - This determines when the simulation will terminate
    - For example, simulation might stop as soon as the n-th customer starts getting service
    - Queueing delay for the first n jobs are calculated